

Numerical Analysis of PageRank

Sparse Computation and Applications to Citation Networks

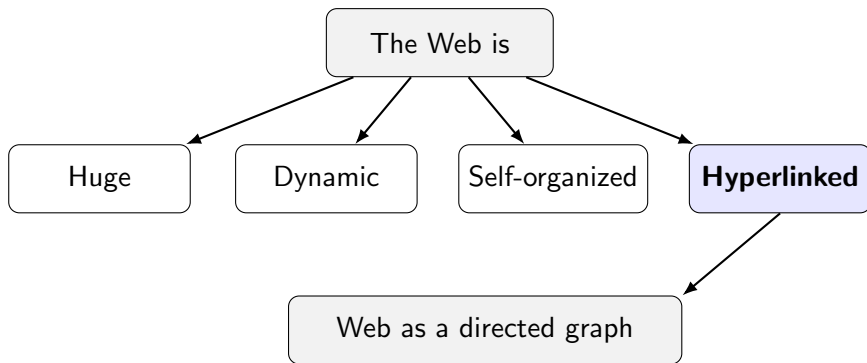
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Why PageRank?

Goal: rank webpages according to their importance



page $i \rightarrow$ page j

a hyperlink from i to j is a recommendation

From Links to the Hyperlink Matrix

From graph to matrix

A directed edge

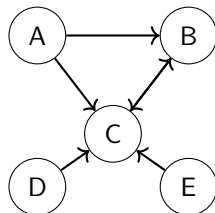
$$i \longrightarrow j$$

means that node i transfers part of its rank to node j .

The hyperlink matrix is defined as

$$H_{ij} = \begin{cases} \frac{1}{d_i}, & \text{if } i \rightarrow j, \\ 0, & \text{otherwise,} \end{cases}$$

where d_i is the out-degree of node i .



$$\mathbf{x}^{(k+1)} = H^T \mathbf{x}^{(k)}$$

The transpose collects the rank received by each node.

Rank propagation

Each node distributes its score equally among its outgoing links.

Why the Raw Matrix Is Not Enough?

Using only the hyperlink matrix, the iteration is

$$\mathbf{x}^{(k+1)} = H^T \mathbf{x}^{(k)}.$$

Dangling nodes

Nodes with no outgoing links produce zero rows in H .

Rank mass can be lost.

Rank sinks

A closed group of nodes can trap the random surfer.

The final ranking may depend on the initial vector.

Cycles

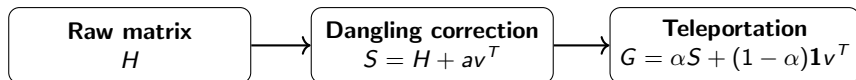
Periodic structures can produce oscillations.

The iteration may fail to converge.

Main point

The raw hyperlink matrix does not always define a unique and convergent PageRank model.

Dangling Correction and Teleportation



Dangling correction

- Dangling nodes have no outgoing links.
- Their missing probability mass is redistributed using v .
- With uniform teleportation:

$$v = \frac{1}{n}\mathbf{1}.$$

Teleportation

- The surfer may jump to any node.
- This prevents traps and disconnected components.
- It makes the final ranking well-defined.

Google Matrix and Convergence

$$G = \alpha S + (1 - \alpha)\mathbf{1}v^T, \quad 0 < \alpha < 1$$

Properties of G

- stochastic;
- positive;
- irreducible;
- aperiodic.

Perron–Frobenius consequence

There exists a unique positive stationary distribution:

$$G^T \mathbf{x} = \mathbf{x}, \quad \sum_i x_i = 1.$$

PageRank vector

The PageRank vector is the stationary distribution of the random surfer defined by the Google matrix.

Role of α : with probability α the surfer follows a link, while with probability $1 - \alpha$ the surfer teleports. Larger values of α give more weight to the graph structure, but usually slow down convergence.

Equivalent Formulations of PageRank

The PageRank vector is the stationary distribution of the Google matrix:

$$G^T \mathbf{x} = \mathbf{x}, \quad \mathbf{x} \geq 0, \quad \sum_i x_i = 1.$$

Fixed point

$$\mathbf{x}^{(k+1)} = G^T \mathbf{x}^{(k)}$$

Power iteration

Eigenvector

$$G^T \mathbf{x} = \mathbf{1} \cdot \mathbf{x}$$

Dominant eigenvector

Linear system

$$(I - \alpha S^T) \mathbf{x} = (1 - \alpha) \mathbf{v}$$

Independent validation

Remark: For large graphs, the power method is preferred because it only requires matrix-vector products.

Sparse Matrix-Free Computation

Problem: teleportation makes the Google matrix G dense.

Dense formulation

$$\mathbf{x}^{(k+1)} = G^T \mathbf{x}^{(k)}$$

- Build the full Google matrix G .
- Matrix-vector product with a dense matrix.

work per iteration: $O(n^2)$

memory: $O(n^2)$

Sparse matrix-free formulation

$$\mathbf{x}^{(k+1)} = \alpha H^T \mathbf{x}^{(k)} + \left(\alpha \mathbf{a}^T \mathbf{x}^{(k)} + 1 - \alpha \right) \mathbf{v}$$

- Store only the sparse hyperlink matrix H .
- Add corrections without forming G .

work per iteration: $O(m + n)$

memory: $O(m + n)$

Power Iteration Workflow

Matrix-free PageRank update

$$\mathbf{x}^{(k+1)} = \underbrace{\alpha H^T \mathbf{x}^{(k)}}_{\text{link contribution}} + \underbrace{\alpha (a^T \mathbf{x}^{(k)}) \mathbf{v}}_{\text{dangling nodes}} + \underbrace{(1 - \alpha) \mathbf{v}}_{\text{teleportation}}$$

One iteration

Step	Computation	Meaning
1	$\mathbf{q}^{(k)} = H^T \mathbf{x}^{(k)}$	link contribution
2	$s^{(k)} = a^T \mathbf{x}^{(k)}$	dangling mass
3	$\mathbf{y} = \alpha \mathbf{q}^{(k)} + (\alpha s^{(k)} + 1 - \alpha) \mathbf{v}$	new vector
4	$r = \ \mathbf{y} - \mathbf{x}^{(k)}\ _1$	residual

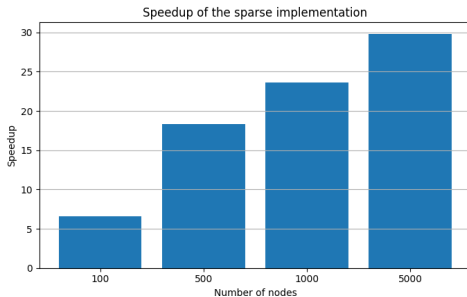
$$r < \varepsilon \Rightarrow \text{stop}, \quad r \geq \varepsilon \Rightarrow \mathbf{x}^{(k+1)} \leftarrow \mathbf{y}.$$

Key point: the dense Google matrix G is never formed explicitly.

Numerical Experiments and Results

Sparse computation · Citation network · Dynamic update

Dense vs Sparse Computation



Experiment

- Random directed graphs.
- Increasing number of nodes.
- Same PageRank model.
- Dense and sparse implementations compared.

Observation

As the graph size increases, the dense formulation becomes dominated by the cost of working with the full Google matrix, while the sparse matrix-free method only processes the existing edges.

Effect of the Damping Factor

$$G = \alpha S + (1 - \alpha) \mathbf{1} v^T$$

The damping factor α controls the balance between following links and teleporting.

Modelling effect

Larger α



more weight to the graph structure

Numerical effect

Larger α

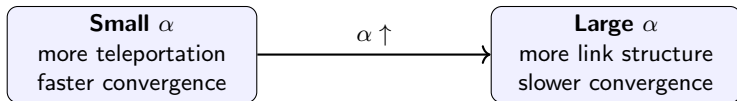
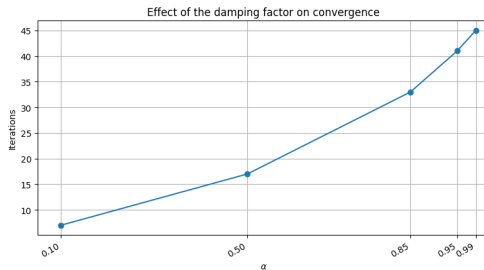


slower convergence of power iteration

Observation

In the experiments, α affected the number of iterations much more than the global ranking.

Damping Factor and Convergence



Observation

Increasing α gives more weight to the graph structure, but it also increases the number of power iterations.

HEP-TH Citation Network

Dataset

- High Energy Physics Theory citation network.
- Distributed by SNAP.
- Nodes represent scientific papers.
- Directed edges represent citations.

Processed graph

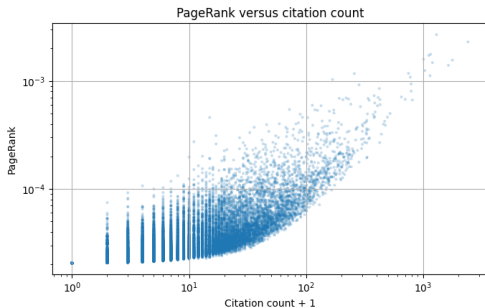
Nodes	27 769
Edges	352 768
Dangling nodes	2 714

Goal

Compare citation count with structural importance measured by PageRank.

$i \longrightarrow j \iff \text{paper } i \text{ cites paper } j$

PageRank vs Citation Count



Comparison

- Citation count is a local measure.
- PageRank is a structural measure.
- The rankings are strongly related, but not identical.

$$\rho_S \approx 0.868$$

Spearman correlation

Observation

Papers with similar citation counts can obtain different PageRank scores, depending on the importance of the papers citing them.

Candidate Scientific Gems

Idea

Citation count may miss papers that are structurally important in the citation network.

A paper is selected when it has:

- high PageRank position;
- much lower citation-count position.

$$R_i = \frac{r_i^{\text{cit}}}{r_i^{\text{PR}}}$$

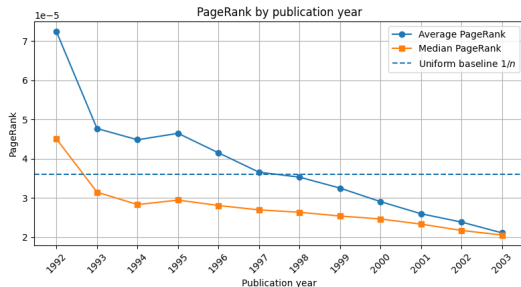
large R_i : better PageRank rank than citation rank

Result

Using top 100 PageRank papers and $R_i > 10$, the experiment identifies 13 candidate scientific gems.

Remark: PageRank measures structural importance, not scientific quality directly.

Effect of Publication Year



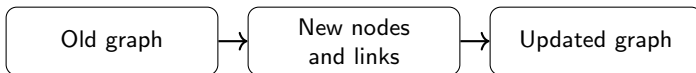
Observed trend

$$\rho_S \approx -0.395$$

- PageRank decreases for more recent papers.
- Recent papers have less time to receive citations.

Remark: this reflects the finite citation window, not an intrinsic preference of PageRank for older papers.

Dynamic PageRank Update



Goal: recompute PageRank on the updated graph more efficiently.

Cold start

Uniform initialization

$$x^{(0)} = \frac{1}{n} \mathbf{1}$$

Warm start

Reuse old PageRank

$$x^{(0)} \approx x_{\text{old}}$$

Aggregation + refinement

Reduced graph initialization

$$x^{(0)} \approx x_{\text{agg}}$$

Observation

Better initializations reduce the number of iterations, while the final PageRank model remains unchanged.

Conclusions

PageRank combines graph modelling with efficient numerical computation.

Mathematical model

Stationary distribution
Eigenvector problem

Computation

Power iteration
Sparse matrix-free form

Applications

Citation networks
Dynamic updates

- Sparse computation avoids constructing the dense Google matrix.
- PageRank captures structural importance beyond citation count.
- Better initializations reduce the cost of dynamic recomputation.

Graph theory + Markov chains + Eigenvectors + Sparse computation

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